In this chapter I'm going to be discussing the first of these points, authentication and membership. In the next chapter I'll tackle the second point, authorization. In section 23.1 I discuss the difference between authentication and authorization, how authentication works in a traditional ASP.NET Core web app, and ways you can architect your system to provide sign-in functionality. I don't discuss API applications in detail in this chapter, though many of the authentication principles apply to both styles of app. I discuss API applications in chapter 25.

In section 23.2 I introduce a user-management system called ASP.NET Core Identity (Identity for short). Identity integrates with Entity Framework Core (EF Core) and provides services for creating and managing users, storing and validating passwords, and signing users in and out of your app.

In section 23.3 you'll create an app using a default template that includes ASP.NET Core Identity out of the box. This gives you an app to explore and see the features Identity provides, as well as everything it doesn't.

Creating an app is great for seeing how the pieces fit together, but you'll often need to add users and authentication to an existing app. In section 23.4 you'll see the steps required to add ASP.NET Core Identity to an existing app.

In sections 23.5 and 23.6 you'll learn how to replace pages from the default Identity UI by scaffolding individual pages. In section 23.5 you'll see how to customize the Razor templates to generate different HTML on the user registration page, and in section 23.6 you'll learn how to customize the logic associated with a Razor Page. You'll see how to store additional information about a user (such as their name or date of birth) and how to provide them permissions that you can later use to customize the app's behavior (if the user is a VIP, for example).

Before we look at the ASP.NET Core Identity system specifically, let's take a look at authentication and authorization in ASP.NET Core—what's happening when you sign in to a website and how you can design your apps to provide this functionality.